Ramsey's theorem for pairs preserves a prescribed ordinal well-foundedness while computing the jump

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Editorial note. The counted unit is the article's own Proposition prop:rt22-wf at source lines 1356-1363 together with the proof environment that begins on the next line (1364-1370); the pairing is the ordinary adjacent one.

Editorial note. The three context spans are the statements of the COH, SEM and SADS propositions that the counted proof applies in sequence; their proofs are not retained and they are counted by sibling cards of this lane. Note that the SADS proposition at source lines 1102-1109 is retained here only as context: as a counted unit that interval is dispositioned, because its own proof environment contains the counted lemma lem:sads-forcing-question-sigma01.

Editorial note. The counted proof carries the author citation of Fiori-Carones et al inside its first line. The author's own bibliography file biblio.bib from the pinned e-print of arXiv:2402.11616v3 is attached to this task unedited and complete, with the bibtex backend, so that the citation resolves.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained.

Editorial note. The article's conservation theorems and its other propositions, lemmas and corollaries are not counted here.

Allowed standard prerequisites (editorial).

- Omega-extension of a topped model into a model of WKL_0-star. The proof starts from a result of Fiori-Carones et al (their Lemma 3.2) that from the auxiliary set P one obtains a P-computable countable coded model of WKL_0-star containing the jump of Y, hence a set P_0 with P Turing above P_0 Turing above the jump of Y; the citation is carried in the proof and its entry is attached with this card.. It supplies the intermediate set P_0 that the first application of the cohesive-set proposition needs.

- The three forcing propositions chained by the proof. The article proves that the COH forcing produces a cohesive set while preserving well-foundedness of a power of alpha and controlling P_0 relative to the jump; that the SEM forcing over that model produces a transitive subtournament with the same properties for one lower power; and that the SADS forcing produces a homogeneous set preserving well-foundedness of alpha itself; their statements are retained with this card at source lines 461-468, 760-767 and 1102-1109.. The proof applies the three propositions one after another to descend from well-foundedness of omega_4 to the fourth power of alpha down to alpha itself, and to obtain the homogeneity of the final set.
- Stable colourings, SEM instances and SADS instances. A colouring on the model is an instance of the cohesiveness principle, and its restriction to a cohesive set is stable; a stable colouring restricted further is an instance of the ascending-descending sequence principle on a transitive set; the article identifies these instances in the proof.. These identifications are what allow the three propositions to be applied in sequence to the same colouring.

Proposition 0.1. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$ topped by a set $Y \in S$, where $\alpha \leqslant \epsilon_0$ is an ordinal. Then for every countable sequence of sets $\vec{R} \in S$ and every $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}^*_0$, there exists $G \subseteq M$ such that*

1. *G is $\vec{R}$-cohesive ;*
2. *$P \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.*

Proposition 0.2. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\sup_{\beta<\alpha} \omega^{\beta\times\omega})$ topped by a set $Y \in S$, where $\alpha \in M$ is a non-zero ordinal $\leqslant \epsilon_0$. Then for every infinite stable tournament R in S and every set $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}^*_0$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite transitive subtournament ;*
2. *$P \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.*

Proposition 0.3. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\sup_{\beta<\alpha} \omega^{\beta\times\omega})$ topped by a set $Y \in S$, where $\alpha \in M$ is an ordinal $\leqslant \epsilon_0$. Then for every linear order $L = (M, \leqslant_L) \in S$ of order type $\omega + \omega^*$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite ascending or descending sequence for $\leqslant_L$;*
2. *$Y' \geqslant_T (G \oplus Y)'$;*

3. $\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.

Proposition 0.4. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\omega_4^\alpha)$ topped by a set $Y \in S$, where $\alpha \in M$ is an ordinal $\leqslant \epsilon_0$. Then, for every coloring $f : [M]^2 \to 2$ in S and every set $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}^*_0$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite f-homogeneous set ;*
2. *$P \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.*

Proof. By Fiori-Carones et al [1, Lemma 3.2], there exists a P-computable countable coded model of WKL^*_0 containing Y'. In particular, there exists a set $P_0 \subseteq M$ such that $P \gg P_0 \gg Y'$. Let $\vec{R} = R_0, R_1, \dots$ be defined by $R_x = \{y : f(x, y) = 1\}$. By Proposition 0.1 applied to $\mathcal{M}$ and P_0, there exists some M-infinite $\vec{R}$-cohesive set $G_0 \subseteq M$ such that $P_0 \geqslant_T (G_0 \oplus Y)'$ and $\mathcal{M}[G_0] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\omega_3^\alpha)$. In particular, $f : [G_0]^2 \to 2$ is stable. One can see $f : [G_0]^2 \to 2$ as an instance of SEM. By Proposition 0.2 applied to $\mathcal{M}[G_0]$ and P, there exists some M-infinite f-transitive set $G_1 \subseteq G_0$ such that $P \geqslant_T (G_1 \oplus G_0 \oplus Y)'$ and $\mathcal{M}[G_1 \oplus G_0] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\omega_2^\alpha)$. One can see $f : [G_1]^2 \to 2$ as an instance of SADS. By Proposition 0.3 applied to $\mathcal{M}[G_0 \oplus G_1]$ and P, there exists some M-infinite f-homogeneous set $G \subseteq G_1$ such that $P \geqslant_T (G \oplus G_1 \oplus G_0 \oplus Y)'$ and $\mathcal{M}[G \oplus G_1 \oplus G_0] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$. □

References

[1] Marta Fiori-Carones, Leszek Aleksander Kołodziejczyk, Tin Lok Wong, and Keita Yokoyama. An isomorphism theorem for models of Weak König's Lemma without primitive recursion. *Journal of the European Mathematical Society*, 2021. To appear.